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A 16GS/s 10b Time-domain ADC using Pipelined-SAR TDC with Delay Variability Compensation and Background Calibration Achieving 153.8dB FoM in 4nm CMOS

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Abstract

The paper presents a direct-RF sampling time-domain ADC achieving 16GS/s 10-bit conversion using merely 4x time-interleaving in 4nm. A redundancy-based TDC delay variability compensation scheme, a background delay offset calibration scheme, and a bottom-plate sampling VTC structure are proposed to achieve high SNR and linearity. The ADC achieves a 55.93dB SFDR and a 44.48dB SNDR at Nyquist with 94.2mW power consumption and 8000um^2 active area, leading to a state-of-the-art 153.8dB Schreier FoM.

Introduction

Direct-RF sampling has gained increasing interest in advanced wireless and wireline communications in recent years due to its simplified receiver front end. Most existing designs implemented time-interleaved (TI) SAR ADCs or pipeline ADCs [1-3]. However, the limited speed of these voltage-domain ADCs inevitably requires excessive time-interleaving, resulting in high power/area consumption and complexity in clock distribution. Recent developments in time-domain ADCs [4-6] have showcased high-speed, high-efficiency ADC channels that are particularly advantageous for advanced technology nodes due to their digital-like delay-based quantization scheme. Nevertheless, existing time-domain ADCs are generally limited in ENOB (<6 bits) due to TDC static non-linearity caused by the variability of delay stages and the distortion caused by the voltage-to-time conversion. To enhance the ENOB while maintaining a high sampling speed, we propose 1) a SAR TDC [6] delay variability compensation and background calibration scheme to break the fundamental static non-linearity limitation; 2) a bottom-plate sampling with top-plate ramp injection voltage-to-time converter (VTC) to significantly reduce distortion.

SAR TDC delay variability compensation via redundancy

Fig. 1 shows the single-stage block diagram of a SAR TDC with the delay variabilities and the proposed compensation via redundant stages. The operation of a SAR TDC is similar to a voltage-domain SAR ADC except that the signal is represented in time. In our implementation, the signal is encoded as the rising edge time difference of two pseudo-differential pulses throughout the TDC stages. A time comparator recognizes which input rising edge arrives earlier, and its output determines whether reference time (T_{ref}) subtraction is applied on the positive or negative path. An extra delay (τ_{comp}) is inserted into both signal paths to hold the time signal while the comparator is resolving. Due to process variation, the delay variability, i.e. (actual delay – nominal delay), of all the delay cells, together with the input-referred comparator offset, can affect the static linearity of the ADC. Separately detecting and calibrating each variability source can cause significant design complexity, jitter, and power/area overhead. To simplify the calibration scheme, we lump all the variabilities into two equivalent error models, including the equivalent time radix error ($\delta_{ref,eq}$) and the equivalent delay offset between the two signal paths ($\delta_{os,eq}$). Both errors can saturate the following stages and cause larger quantization noise. To compensate for those errors, we propose to add a few redundant stages at the end of the SAR TDC conversion chain. As long as the additional quantization range provided by redundant stages covers the variability-induced errors, the final quantization noise can be reduced to within 1 LSB of the target specification.

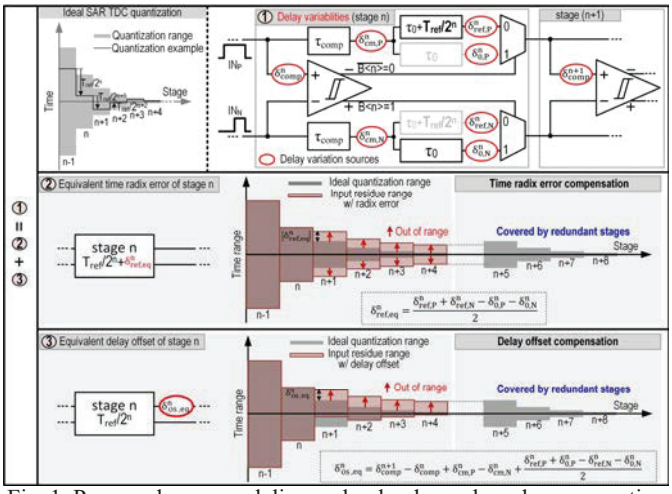


Fig. 1. Proposed error modeling and redundancy-based compensation for a SAR TDC.

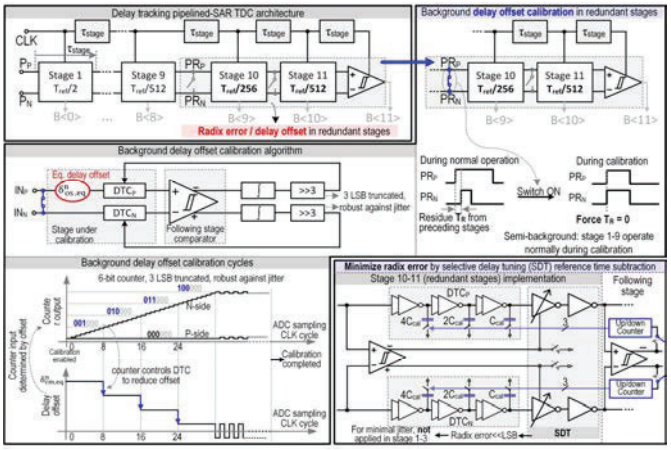


Fig. 2. Proposed background delay offset calibration for the redundant stages of the delay-tracking pipelined-SAR TDC.

Background delay offset calibration in redundant stages

Fig. 2 shows the overall TDC architecture using delay-tracking pipelined-SAR TDC [6] and the proposed background delay offset calibration scheme to ensure the accuracy of the redundant stages. Since the redundant stages are inserted at the end of the conversion, their variabilities cannot be further compensated and may degrade the effectiveness of error correction as outlined in Fig. 1. Therefore, the variabilities need to be minimized by either design or calibration. For the time radix error of each redundant stage, we apply the selective

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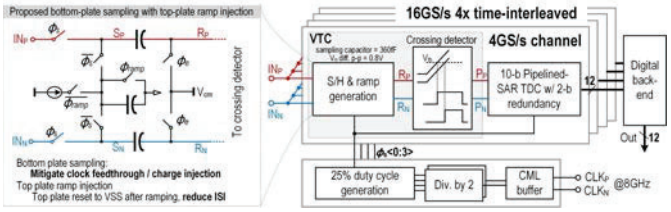


Fig. 3. ADC block diagram and the proposed VTC architecture.

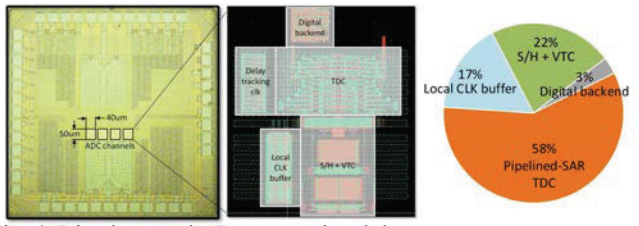


Fig. 4. Die photo and ADC power breakdown.

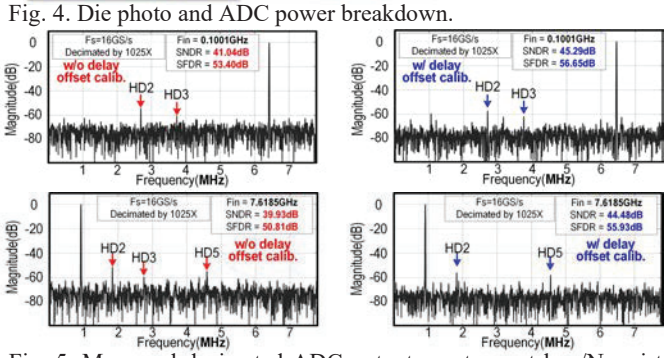


Fig. 5. Measured decimated ADC output spectrum at low/Nyquist input frequency with and without the delay offset calibration.

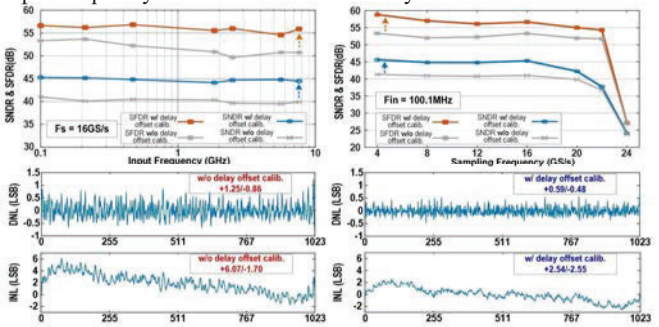
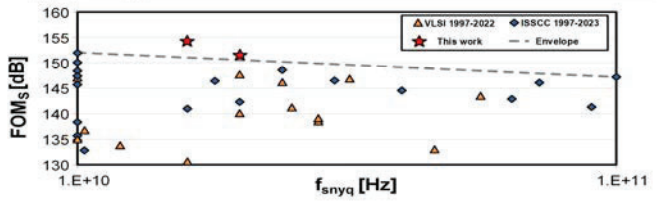


Fig. 6. Measured SNDR/SFDR versus input/sampling frequencies, and INL/DNL with and without the delay offset calibration.

Table I. Performance summary and comparison table

	This work		M. Zhang VLSI 2021	J. Liu ISSCC 2022	A. M. A. Ali ISSCC 2020	S. S. Kumar ISSCC 2023	L. Kull VLSI 2018
Architecture	TI-Time-Domain				TI-Pipeline	TI-SAR	
Technology	4nm FinFET		65nm CMOS	12nm FinFET	16nm FinFET	7nm FinFET	14nm FinFET
Sampling Speed [GS/s]	16	20	20	10	18	24	32
TI-channels	4x	4x	8x	2x	8X	80x	48x
Resolution [b]	10	10	8	8	12	12	10
Supply voltage [V]	0.9/0.75*	0.9/0.75*	1	0.8	N/A	N/A	0.95/0.8*
SNDR [dB]	44.5	42.1	38.8	37.2	48	46.5	37.8
SFDR [dB]	55.9	55.0	52.5	50.69	54	76	48.21
Power [mW]	94.2	116.7	129.9	14.8	1300	750	199
Area [mm^2]	0.008	0.008	0.223	0.00285	2.6	0.9	0.16
NSD@Nyquist [dBFS/Hz]	143.5	142.1	138.8	134.2	147.5	147.3	139.8
FoM$_{Schreier}$[dB]	153.8	151.4	147.6	152.5	146.4	148.5	146.9

*S/H supply / quantizer supply



delay tuning (SDT) technique [6] for reference time subtraction to minimize the error (<<LSB) by design. As for the delay offset of the redundant stages, we propose to calibrate it in the background. During calibration, the input rising edges of the redundant stage under calibration are forced to be aligned through a shorting switch while the first 9 SAR TDC stages still operate normally. In this case, the following stage comparator output is solely determined by the delay offset of the stage under calibration. Accordingly, we apply two integrators to accumulate the comparator's positive/negative outputs individually and adjust the corresponding DTC to compensate for the delay offset. To improve the offset estimation robustness against jitter, the integrator output LSBs are truncated. After sufficient cycles (usually <64), the delay difference between the positive and negative paths will be minimized. At that point, the calibration for the current redundant stage is completed, and ready for the calibration of the next redundant stage. As shown later by the measurement results, this calibration scheme can reduce the ADC static nonlinearity by ~3X.

ADC architecture and VTC circuit implementation

Fig. 3 shows the top-level ADC architecture. The overall ADC achieves 16GS/s using only 4-way time-interleaving. In each channel, the sampled input differential voltage is linearly converted into a time difference via a VTC, and further quantized by the aforementioned pipelined-SAR TDC. Compared to the two-step Flash-SAR TDC used in [6], we eliminate the first-step Flash TDC to avoid its significant delay variability and high static non-linearity.

To improve the VTC linearity, we propose the bottom-plate sampling with top-plate ramp injection, as shown in Fig. 3. In the proposed VTC, we first use bottom plate sampling to mitigate the clock feedthrough and charge injection. At the end of the sampling phase, the sampled voltage is ramped up by injecting a DC current to the top plates, such that the voltage at the bottom plates will cross a certain voltage level (V_{th}) which effectively encodes the sampled voltage into the time difference between the two crossing points, as shown in Fig. 3. More importantly, when the ramp voltage is reset before the next sampling phase, it eliminates both common-mode and differential-mode kickbacks to the input and hence reducing the ISI. The SPICE simulation indicates a 6.5dB SFDR improvement compared to the conventional VTC structure.

Measurement Results

The ADC is fabricated in 4nm CMOS with an active area of 8000um^2, as shown in Fig. 4. Fig. 5 shows the measured spectrum of the decimated ADC output, with the time-interleaving clock skew calibrated on chip using tunable delay lines, and inter-channel gain errors and offsets calibrated off chip. The ADC achieves a 45.3dB SNDR and a 56.7dB SFDR at low input frequency, and a 44.5dB SNDR and 55.9dB SFDR at Nyquist. Fig. 6 shows that the 16GS/s ADC achieves >44.3dB SNDR and >55dB SFDR across the input frequency from 100MHz to 7.62GHz. At a sampling speed of 20GS/s, the ADC still achieves SNDR of >42dB. The measured static non-linearity shows the DNL within +0.59/-0.48 and INL within +2.54/-2.55 LSBs after the background delay offset calibration in the redundant stages. The ADC consumes 94.2mW power at 16GS/s, leading to a record-high 153.8dB Schreier FoM. Table I shows that this ADC achieves much higher SNDR compared to other time-domain ADCs, and much lower power and area consumption compared to conventional voltage-domain ADCs.

References

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